

Lecture 4

Density Estimation and Gaussian Mixture Models

Introducing Maximum Likelihood Estimation, and Gaussian Mixture Models

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Roadmap

- Expectation and Variance
- Density Estimation
 - Basic Probability Distributions
 - Maximum Likelihood
 - Bias
- Multivariate Normal
- Gaussian Mixture Models

Questions

Expectation and Variance

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- Density Estimation
 - Basic Probability Distributions
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Questions

Expectations

The **expectation of a random variable** X is the the weighted average with respect to a probability distribution $p(x)$:

$$\mathbb{E}[X] = \sum_x x p(x)$$

Expectation is an **operator** from a **random variable** to a *value*.

The **conditional expectation** of a random variable is a **weighted average** with respect to the **conditional distribution**:

$$\mathbb{E}[X \mid y] = \sum_x x p(x|y)$$

Functions of Random Variables

The function of a random variable is also a random variable:

$$Y = f(X)$$

The **expectation of a function** $f(X)$ is the weighted average with respect to a probability distribution $p(x)$:

$$\mathbb{E}[f] = \mathbb{E}[f(X)] = \sum_x f(x) p(x)$$

The expectation of a function of multiple variables is defined over the joint distribution:

$$\mathbb{E}[f(X, Y)] = \sum_{x, y} f(x, y) p(x, y)$$

Linearity of Expectation

For *any* two random variables X and Y :

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

- Does not require any assumptions about $p(X, Y)$.

$$\begin{aligned}\mathbb{E}[aX + bY + c] &= \sum_x \sum_y (a x + b y + c) p(x, y) \\&= \sum_x \sum_y (a x p(x, y) + b y p(x, y) + c p(x, y)) \\&= a \sum_x x \sum_y p(x, y) + b \sum_y y \sum_x p(x, y) + c \sum_x \sum_y p(x, y) \\&= a \mathbb{E}[X] + b \mathbb{E}[Y] + c\end{aligned}$$

Variance

The **variance** of a random variable defines its **spread**:

$$\text{var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Using linearity of expectation, we can rewrite the variance as:

$$\text{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Variance

1. 从定义出发

方差的原始定义是随机变量 X 偏离其均值 $E[X]$ 的平方的期望：

$$\text{var}[X] = E[(X - E[X])^2]$$

2. 展开平方项

根据代数公式 $(a - b)^2 = a^2 - 2ab + b^2$ ，我们将括号内展开：

$$\text{var}[X] = E[X^2 - 2X \cdot E[X] + (E[X])^2]$$

3. 利用“期望的线性性质”

这是最关键的一步。期望算子 $E[\cdot]$ 具有线性性质，这意味着：

- 和的期望等于期望的和： $E[A + B] = E[A] + E[B]$
- 常数项可以提取： $E[aX] = aE[X]$ （注意： $E[X]$ 在这一步中被视为一个常数，因为它是一个确定的数值）。

Variance

我们将 $E[\cdot]$ 分配到每一项中：

$$\text{var}[X] = E[X^2] - E[2X \cdot E[X]] + E[(E[X])^2]$$

4. 化简

- 中间项： $2E[X]$ 是常数，提取出来后变为： $2E[X] \cdot E[X] = 2(E[X])^2$
- 最后一项： $(E[X])^2$ 本身就是个常数，它的期望还是它自己： $(E[X])^2$

代回原式：

$$\text{var}[X] = E[X^2] - 2(E[X])^2 + (E[X])^2$$

5. 合并同类项

$$-2(E[X])^2 + (E[X])^2 = -(E[X])^2$$

最终得出：

$$\text{var}[X] = E[X^2] - (E[X])^2$$

Variance

Using linearity of expectation, we can rewrite the variance as:

$$\text{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

What is $\text{var}[aX + b]$?

$$\text{var}[aX + b] = \underbrace{\mathbb{E}[(aX + b)^2]} - \underbrace{\mathbb{E}[aX + b]^2}$$

$$\mathbb{E}[a^2X^2 + 2abX + b^2]$$

$$- (a\mathbb{E}[X] + b)^2$$

$$a^2\mathbb{E}[X^2] + \cancel{2ab\mathbb{E}[X]} + \cancel{b^2}$$

$$- a^2\mathbb{E}[X]^2 - \cancel{2ab\mathbb{E}[X]} - \cancel{b^2}$$

$$\text{var}[aX + b] = a^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) = a^2 \text{var}[X]$$

Covariance

The covariance of two random variables defines the degree to which they vary together:

$$\text{cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

1. 从定义出发

根据协方差的定义：

$$\text{cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$$

2. 展开乘积项

利用乘法分配律（FOIL法则：First, Outer, Inner, Last），展开括号内的项：

$$(X - E[X])(Y - E[Y]) = XY - X \cdot E[Y] - Y \cdot E[X] + E[X]E[Y]$$

3. 应用期望算子 $E[\cdot]$

利用期望的线性性质，将期望算子作用于每一项。注意：在此过程中， $E[X]$ 和 $E[Y]$ 是常数，可以直接提取：

$$\text{cov}[X, Y] = E[XY - X \cdot E[Y] - Y \cdot E[X] + E[X]E[Y]]$$

拆开为：

$$\text{cov}[X, Y] = E[XY] - E[X \cdot E[Y]] - E[Y \cdot E[X]] + E[E[X]E[Y]]$$

4. 提取常数项进行化简

由于 $E[X]$ 和 $E[Y]$ 已经是对 X 和 Y 求过期望后的数值（常数），所以：

- $E[X \cdot E[Y]] = E[Y] \cdot E[X]$
- $E[Y \cdot E[X]] = E[X] \cdot E[Y]$
- $E[E[X]E[Y]] = E[X]E[Y]$ （常数的期望就是常数本身）

代入式子中：

$$\text{cov}[X, Y] = E[XY] - E[Y]E[X] - E[X]E[Y] + E[X]E[Y]$$

5. 合并同类项

$$\text{cov}[X, Y] = E[XY] - 2E[X]E[Y] + E[X]E[Y]$$

$$\text{cov}[X, Y] = E[XY] - E[X]E[Y]$$

Covariance

The covariance of two **random vectors** $\mathbf{x}$ and $\mathbf{y}$ is the matrix:

$$\text{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{y}^T - \mathbb{E}[\mathbf{y}^T])] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$$

推导步骤

我们将协方差的定义式展开：

$$\text{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{y}^T - \mathbb{E}[\mathbf{y}^T])]$$

第一步：展开括号中的乘法

利用矩阵乘法的分配律，将项展开：

$$(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{y}^T - \mathbb{E}[\mathbf{y}^T]) = \mathbf{x}\mathbf{y}^T - \mathbf{x}\mathbb{E}[\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbf{y}^T + \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$$

第二步：利用期望的线性性质

将 $\mathbb{E}$ 作用于展开后的每一项：

$$\mathbb{E}[\dots] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}\mathbb{E}[\mathbf{y}^T]] - \mathbb{E}[\mathbb{E}[\mathbf{x}]\mathbf{y}^T] + \mathbb{E}[\mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]]$$

第三步：简化各项

这里是关键点，请注意： $\mathbb{E}[\mathbf{x}]$ 和 $\mathbb{E}[\mathbf{y}^T]$ 本身已经是**常数向量**（标量），对于期望算子而言，它们可以被提取出来：

- 对于第二项： $\mathbb{E}[\mathbf{x}\mathbb{E}[\mathbf{y}^T]] = \mathbb{E}[\mathbf{x}] \cdot \mathbb{E}[\mathbf{y}^T] = \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$
- 对于第三项： $\mathbb{E}[\mathbb{E}[\mathbf{x}]\mathbf{y}^T] = \mathbb{E}[\mathbf{x}] \cdot \mathbb{E}[\mathbf{y}^T] = \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$
- 对于第四项：因为 $\mathbb{E}[\mathbf{x}]$ 和 $\mathbb{E}[\mathbf{y}^T]$ 都是常数，所以它们的期望就是它们本身：
 $\mathbb{E}[\mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]] = \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$

第四步：代入汇总

将简化后的项带回原式：

$$\text{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T] + \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$$

最后，抵消掉多余的项：

$$\text{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$$

Probability Density Functions

For **continuous random variables**, we use a **density function** $p(x)$ where the probability that x falls in the interval (a, b) is given by:

$$p(x \in (a, b)) = \int_a^b p(x) dx$$

Just as with discrete distribution, we have the following two properties:

$$p(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} p(x) dx = 1$$

We define the **cumulative distribution function**:

$$p(X \leq z) = P(z) = \int_{-\infty}^z p(x) dx$$

Density Estimation

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Questions

Density Estimation

Density estimation is the process of trying to **infer the probability distribution $p(X)$** given observations (data) $\mathcal{D} = \{x_1, \dots, x_N\}$.

There are *infinitely many* density functions from which our data could have been sampled.

- Could be any function that assigns **non-zero probability to the data**.
- The choice of function is a **modeling decision (inductive bias!)**.

Density estimation is a fundamental technique in machine learning and is part of most modern methods.

Estimating the Parameters of a Distribution

Given a collection of data $\mathcal{D} = \{x_1, \dots, x_N\}$, there are several ways to estimate the “best” parameters for a given distribution.

- **Maximum Likelihood Estimation (MLE):** Choose the parameters that maximize the **likelihood** of the data. (Frequentist)

$$w_{\text{MLE}}^* = \arg \max_w p(\mathcal{D} \mid w)$$

- **Maximum A Posteriori (MAP) Estimation:** Choose the parameters that maximize the posterior given the data and the **prior**. (Bayesian)

$$w_{\text{MAP}}^* = \arg \max_w p(\mathcal{D} \mid w) p(w)$$

Both methods rely on computing the **likelihood of the data**.

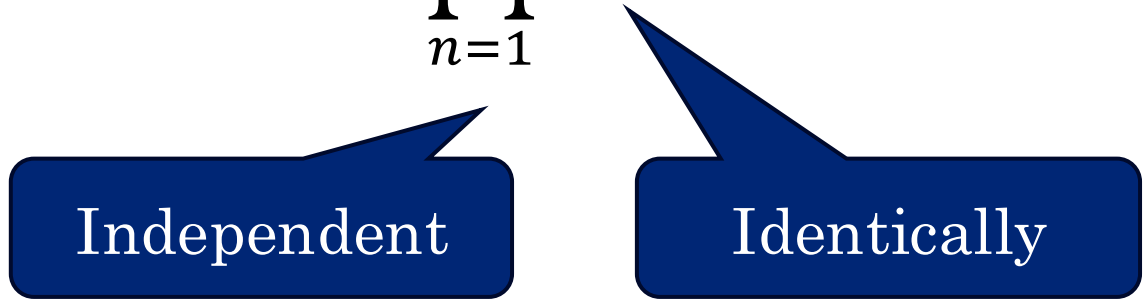
- We will focus on MLE.

The Likelihood Function

The likelihood function describes the **probability of the observed data** $\mathcal{D} = \{x_1, \dots, x_N\}$ under a particular distribution.

If we assume each observation $X_n \sim p(x|w)$ is an **Independent and Identically Distributed (IID)** sample from some parametric distribution $p(x|w)$ parameterized by w , then the likelihood is:

$$p(\mathcal{D}|w) = \prod_{n=1}^N p(x_n | w)$$



Independent

Identically

The Log Likelihood Function

In practice, we will typically work with the **log likelihood function**:

$$p(\mathcal{D}|w) = \prod_{n=1}^N p(x_n | w) \quad \Rightarrow \quad \ln p(\mathcal{D}|w) = \sum_{n=1}^N \ln p(x_n | w)$$

We take the **log** of the likelihood function for several reasons: $\frac{d}{dx} \sum_{i=1}^n f_i(x) = \sum_{i=1}^n \frac{d}{dx} f_i(x)$

- It's **easier to manipulate summations** than products (e.g., calculus).
- The log probabilities and summation are **more numerically stable**.

Does this change our **optimization problem**?

- No – log is **monotonically increasing function** of its arguments:

$$\arg \max_w p(\mathcal{D}|w) = \arg \max_w \ln p(\mathcal{D}|w)$$

Maximum Likelihood Estimation

Maximum likelihood estimation has two steps:

1. **Modeling:** Define the **log likelihood function** of the **data**: $\ln p(\mathcal{D}|w)$
2. **Optimization:** Compute the parameters that maximize the log-likelihood:

$$\theta_{\text{ML}} = \arg \max_{\theta} \log p(\mathcal{D}|w)$$

Maximum likelihood estimation is a core technique in ML and is the basic objective for everything from the **logistic regression** to **ChatGPT**.

Modeling: Bernoulli Distribution



The distribution of binary random variable $X \in \{0,1\}$ defined by:

$$X \sim \text{Bern}(x|\mu) \implies p(x|\mu) = \mu^x(1-\mu)^{1-x}$$

where the parameter $0 \leq \mu \leq 1$.

$$\mathbb{E}[X] = 0(1-\mu) + 1\mu = \mu$$

$$\text{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mu - \mu^2 = \mu(1-\mu)$$

$$\mathbb{E}[X^2] = 0^2(1-\mu) + 1^2\mu = \mu$$

Note that for $\mu \notin \{0,1\}$, X never takes the expected value!

The Bernoulli distribution is used extensively in classification.

The MLE for IID Bernoulli Samples (Part 1)

Example: Bernoulli Distribution

We observe N samples $\mathcal{D} = \{x_1, \dots, x_N\}$ from a Bernoulli Distribution.

1. Modeling: Let's assume IID Bernoulli trials:

$$\ln p(\mathcal{D}|\mu) = \sum_{n=1}^N \ln(\mu^{x_n}(1-\mu)^{(1-x_n)}) = \sum_{n=1}^N (x_n \ln(\mu) + (1-x_n) \ln(1-\mu))$$

Since each x_n is either **zero** or **one** we can rewrite this expression as:

$$\ln p(\mathcal{D}|\mu) = n_1 \ln(\mu) + n_0 \ln(1-\mu)$$

Where n_0 and n_1 are the number of observed zeros and ones, respectively.

- The counts n_0 and n_1 are called **sufficient statistics (充分统计量)** because they **fully describe the data**.

The MLE for IID Bernoulli Samples (Part 2)

2. Optimization: Maximize the log likelihood function:

$$\arg \max_{\mu} n_1 \ln(\mu) + n_0 \ln(1 - \mu)$$

Here we use the **critical-points** (临界点) approach to optimization:

1. Take the derivative $\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu)$:

$$\frac{\partial}{\partial \mu} (n_1 \ln(\mu) + n_0 \ln(1 - \mu)) = \frac{n_1}{\mu} - \frac{n_0}{1 - \mu}$$

2. Solve for the critical point $\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu) = 0$

$$\frac{n_1}{\mu} - \frac{n_0}{1 - \mu} = 0 \implies n_1(1 - \mu) = n_0\mu \implies \mu_{ML} = \frac{n_1}{n_0 + n_1}$$

3. Verify that critical points are maxima $\frac{\partial^2}{\partial \mu^2} \ln p(\mathcal{D}|\mu) < 0$

The MLE for IID Bernoulli Samples (Part 3)

2. Optimization: Maximize the log likelihood function:

$$\arg \max_{\mu} n_1 \ln(\mu) + n_0 \ln(1 - \mu)$$

Here we use the analytic **critical-point** based approach to optimization:

1. Take the derivative $\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu) = \frac{n_1}{\mu} - \frac{n_0}{1-\mu}$
2. Solve for the critical points $\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu) = 0 \Rightarrow \mu_{ML} = \frac{n_1}{n_0+n_1}$
3. Verify that critical points are maxima $\frac{\partial^2}{\partial \mu^2} \ln p(\mathcal{D}|\mu) < 0$

$$\frac{\partial^2}{\partial \mu^2} \ln p(\mathcal{D}|\mu) = -\frac{n_1}{\mu^2} - \frac{n_0}{(1-\mu)^2}$$

Because n_0 , n_1 , μ^2 , and $(1-\mu)^2$ are all non-negative:

$$\frac{\partial^2}{\partial \mu^2} \ln p(\mathcal{D}|\mu) = -\frac{n_1}{\mu^2} - \frac{n_0}{(1-\mu)^2} < 0$$



How can we mathematically formulate the problem of next-token prediction as a Maximum Likelihood Estimation (MLE) problem?

The MLE for ChatGPT

We model the likelihood of all the documents on the internet:

$$\begin{aligned} & \ln p(\text{Internet} \mid w) \\ &= \sum_{\text{Doc} \in \text{Internet}} \sum_{\text{Word} \in \text{Doc}} \ln p(\text{Word} \mid \text{Words before it in Doc}, w) \end{aligned}$$

Pre-Training: billions of dollars spent annually computing an MLE:

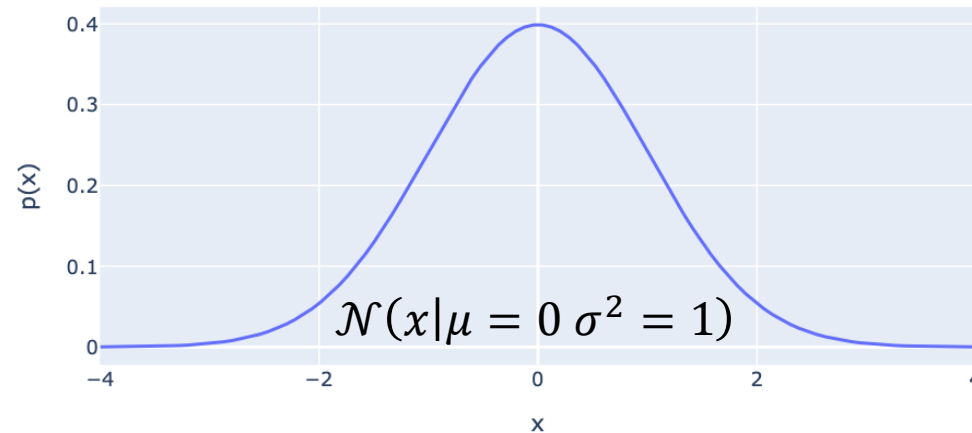
$$\arg \max_w \ln p(\text{Internet} \mid w)$$

Normal (Gaussian) Distribution

If a single real-valued variable X is **normally distributed**, then:

$$X \sim \mathcal{N}(x|\mu, \sigma^2) \quad \Rightarrow \quad p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

Where the parameters μ and σ^2 are called the mean and variance.
 $\mathbb{E}[X] = \mu$ and $\text{var}[X] = \sigma^2$



The MLE for IID Gaussian Samples (Part 1)

Example: Gaussian Distribution

We observe N continuous real valued samples $\mathcal{D} = \{x_1, \dots, x_n\}$.

1. Modeling: Let's assume IID Gaussian trials:

$$\begin{aligned}\ln p(\mathcal{D}|\mu, \sigma^2) &= \sum_{n=1}^N \ln \left(\frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{1}{2\sigma^2} (x_n - \mu)^2 \right) \right) \\ &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2\end{aligned}$$

2. Optimization: Maximize the log likelihood function:

$$\arg \max_{\mu, \sigma^2} -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2$$

The MLE for IID Gaussian Samples (Part 2)

Maximize the log likelihood function:

$$\arg \max_{\mu, \sigma^2} -\cancel{\frac{N}{2} \ln(2\pi)} - \frac{N}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2$$

Compute derivatives with respect to μ and σ^2 and solve for the critical-points.

$$\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu, \sigma^2) = \frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu)$$

Solving for μ_{ML} :

$$\frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu) = 0 \implies \mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n$$

The MLE for IID Gaussian Samples (Part 2)

Computing the derivative with respect to the variance parameter $v = \sigma^2$

$$\frac{\partial}{\partial v} \ln p(\mathcal{D}|\mu, v) = \frac{\partial}{\partial v} \left(-\frac{N}{2} \ln(v) - \frac{1}{2v} \sum_{n=1}^N (x_n - \mu)^2 \right) = -\frac{N}{2} \left(\frac{1}{v} \right) + \frac{1}{2v^2} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2$$

Set equal to zero and solving for $v_{\text{ML}} = \sigma_{\text{ML}}^2$:

$$\begin{aligned} -\frac{N}{2} \left(\frac{1}{v} \right) + \frac{1}{2v^2} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2 &= 0 \\ \frac{v^2}{N} * N \left(\frac{1}{v} \right) &= \frac{1}{v^2} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2 * \frac{v^2}{N} \\ v_{\text{ML}} = \sigma_{\text{ML}}^2 &= \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2 \end{aligned}$$

Recap of the MLE for IID Gaussian Samples

We used the method of **Maximum Likelihood** to estimate the mean and variance parameters of a Gaussian from data:

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n$$

Sufficient Statistics

$$\sigma_{\text{ML}}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2 = \frac{1}{N} \left(\sum_{n=1}^N x_n^2 \right) - \mu_{\text{ML}}^2$$

What are the sufficient statistics?

Do they match our intuition for the mean and variance.

Are they good estimates?

Bias

- Expectation and Variance
- Density Estimation
 - Basic Probability Distributions
 - Maximum Likelihood
 - Bias
- Multivariate Normal
- Gaussian Mixture Models

Questions

Bias of an Estimate

How close is the true (unknown) parameter value to our estimate in expectation?

Before we observe our data, the value of our estimate θ_{ML} is a **random variable**.

We can compute the bias of the estimation procedure as:

$$\text{Bias}(\theta_{\text{ML}}) = \mathbb{E}[\theta_{\text{ML}}] - \theta$$

Relative to the true (but unknown) θ .

Bias of the MLE for the Gaussian Distribution

Assuming the data are drawn IID from $\mathcal{N}(x|\mu, \sigma^2)$ treating X_n , μ_{ML} , and σ_{ML}^2 as random variables

$$\mathbb{E}[\mu_{\text{ML}}] = \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^N X_n\right] = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[X_n] = \frac{1}{N} \sum_{n=1}^N \mu = \mu \quad \text{Unbiased}$$

Now looking at the variance:

$$\begin{aligned} \mathbb{E}[\sigma_{\text{ML}}^2] &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[X_n^2] - \mathbb{E}[\mu_{\text{ML}}^2] \\ &= \frac{1}{N} \sum_{n=1}^N (\sigma^2 + \mu^2) - (\underbrace{\text{var}(\mu_{\text{ML}})}_{\text{var}(\mu_{\text{ML}})} + \underbrace{\mathbb{E}[\mu_{\text{ML}}^2]}_{\mu^2}) \end{aligned}$$

Variance Identity

$$\begin{aligned} \text{var}(X_n) &= \mathbb{E}[X_n^2] - \mu^2 = \sigma^2 \\ \text{var}(\mu_{\text{ML}}) &= \mathbb{E}[\mu_{\text{ML}}^2] - \mathbb{E}[\mu_{\text{ML}}]^2 \end{aligned}$$

$$\text{var}(\mu_{\text{ML}}) = \text{var}\left(\frac{1}{N} \sum_{n=1}^N X_n\right) = \frac{1}{N^2} \underbrace{\left(\sum_{n=1}^N \text{var}(X_n)\right)}_{\text{Independence of } X_n} = \frac{1}{N^2} \underbrace{N \sigma^2}_{\text{var}(X_n)} = \frac{\sigma^2}{N}$$

Bias of the MLE for the Gaussian Distribution

Assuming the data are drawn IID from $\mathcal{N}(x|\mu, \sigma^2)$ treating X_n , μ_{ML} , and σ_{ML}^2 as random variables

$$\mathbb{E}[\mu_{\text{ML}}] = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[X_n] = \frac{1}{N} \sum_{n=1}^N \mu = \mu \quad \text{Unbiased}$$

Now looking at the variance:

$$\begin{aligned} \mathbb{E}[\sigma_{\text{ML}}^2] &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[X_n^2] - \mathbb{E}[\mu_{\text{ML}}^2] \\ &= \frac{1}{N} \sum_{n=1}^N (\sigma^2 + \mu^2) - (\text{var}(\mu_{\text{ML}}) + \mathbb{E}[\mu_{\text{ML}}]^2) \\ &= (\sigma^2 + \mu^2) - \left(\frac{\sigma^2}{N} + \mu^2 \right) = \sigma^2 \left(\frac{N-1}{N} \right) \end{aligned}$$

Biased

Variance Identity

$$\begin{aligned} \text{var}(X_n) &= \mathbb{E}[X_n^2] - \mu^2 = \sigma^2 \\ \text{var}(\mu_{\text{ML}}) &= \mathbb{E}[\mu_{\text{ML}}^2] - \mathbb{E}[\mu_{\text{ML}}]^2 \end{aligned}$$

Sample Mean Variance

$$\text{var}(\mu_{\text{ML}}) = \frac{\sigma^2}{N}$$



How do we derive the formula for an unbiased estimator of the Gaussian variance?

无偏样本方差 s^2 与 MLE 估计 σ_{ML}^2 之间的转换关系为：

$$s^2 = \frac{N}{N-1} \sigma_{\text{ML}}^2$$

这种从 N 到 $N - 1$ 的转变，在统计学上被称为 **Bessel 修正 (Bessel's correction)**，它是从“以数据为中心”推向“以总体分布为中心”的关键一步。

Multivariate Normal

- Expectation and Variance
- Density Estimation
 - Basic Probability Distributions
 - Maximum Likelihood
 - Bias
- Multivariate Normal
- Gaussian Mixture Models

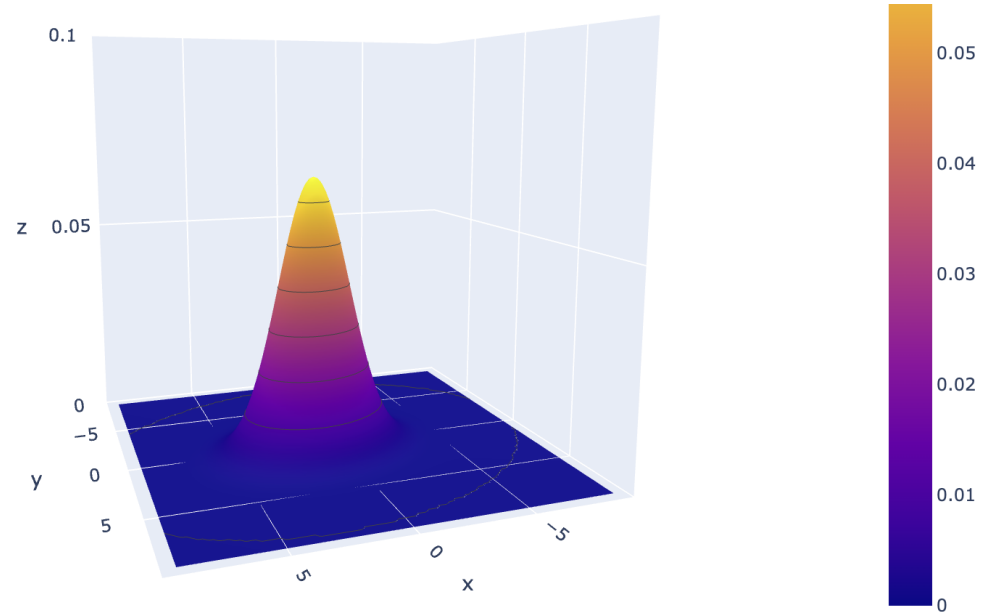
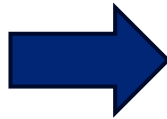
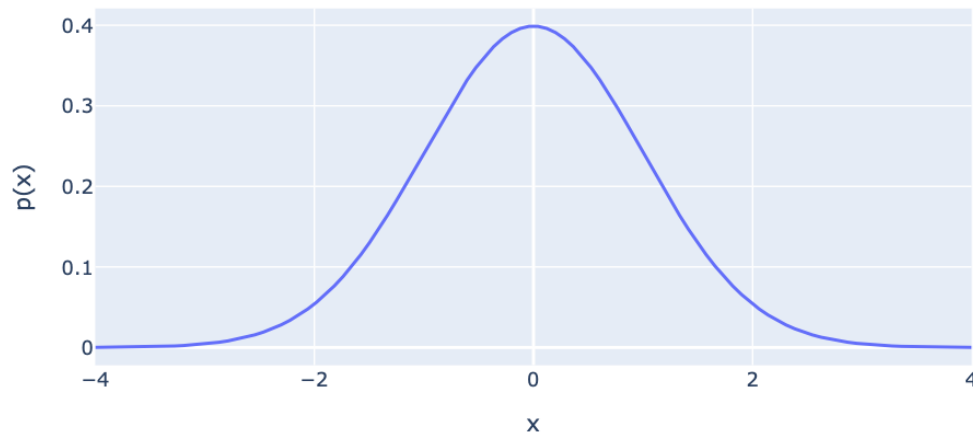
Questions

Multivariate Gaussian Distribution

The **probability density function** for a multivariate Gaussian distribution in D dimensions is give by:

$$\mathcal{N}(\mathbf{x} | \mu, \Sigma) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$$

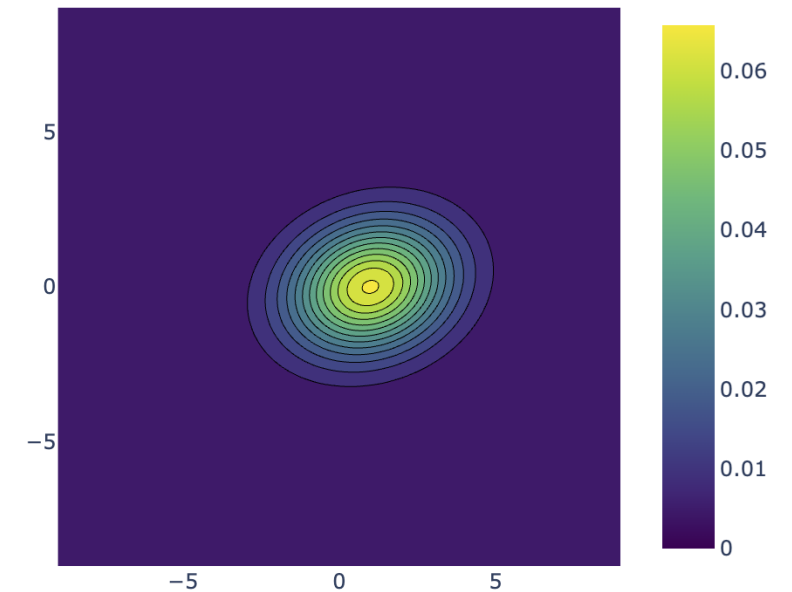
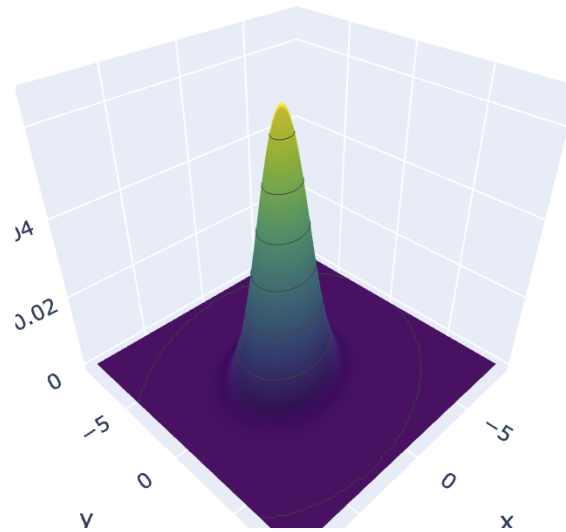
where the **parameters** $\mu \in \mathbb{R}^D$ and $\Sigma \in \mathbb{R}^{D \times D}$ are the **mean vector** and **covariance matrix**.



Demo

Gaussians

```
mu = np.array([1, 0])  
Sigma = np.array([[3, 0.4], [0.4, 2]])  
  
from scipy.stats import multivariate_normal  
normal = multivariate_normal(mean=mu, cov=Sigma)  
normal.pdf(np.array([1, 0.5]))
```



Gaussian Mixture Models

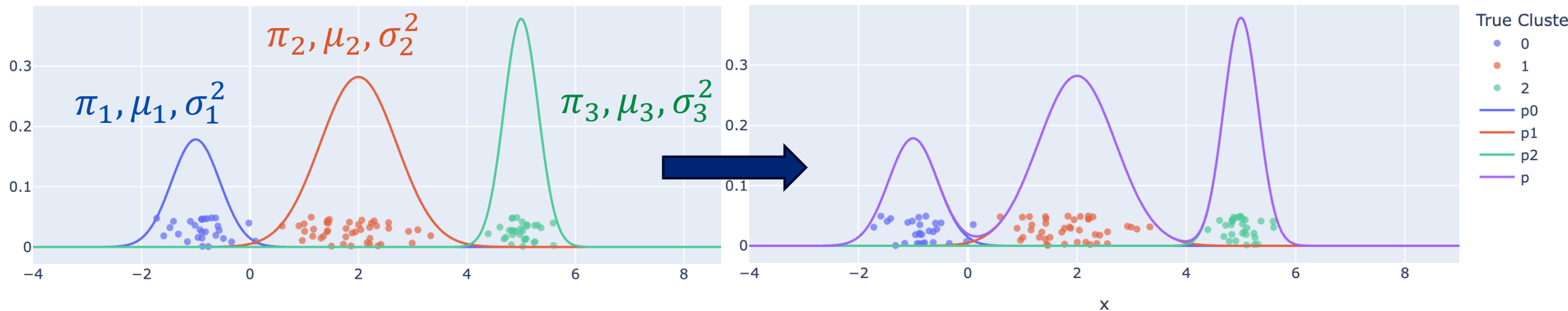
- Expectation and Variance
- Density Estimation
 - Basic Probability Distributions
 - Maximum Likelihood
 - Bias
- Multivariate Normal
- Gaussian Mixture Models

Questions

Gaussian Mixture Model (GMM)

The Gaussian mixture model defines the probability of sampling a D -dimensional vector $\mathbf{x}$ as a **weighted combination** of Gaussians:

- **Example:** $K=3$



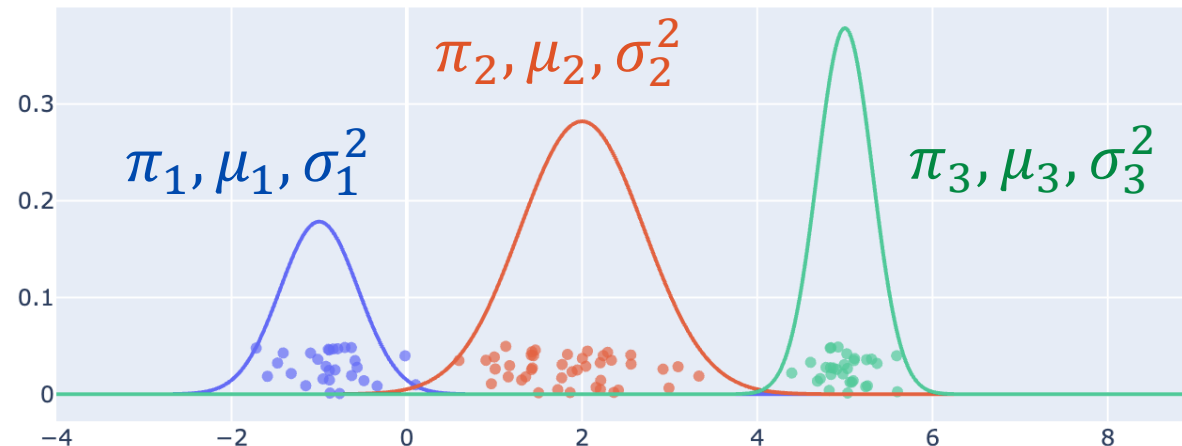
Gaussian Mixture Model (GMM)

The Gaussian mixture model defines the probability of sampling a D -dimensional vector $\mathbf{x}$ as a **weighted combination** of Gaussians:

$$p(\mathbf{x} \mid \pi, \mu_1, \Sigma_1, \dots, \mu_K, \Sigma_K) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k)$$

where $\sum_{k=1}^K \pi_k = 1$ is a probability distribution over clusters.

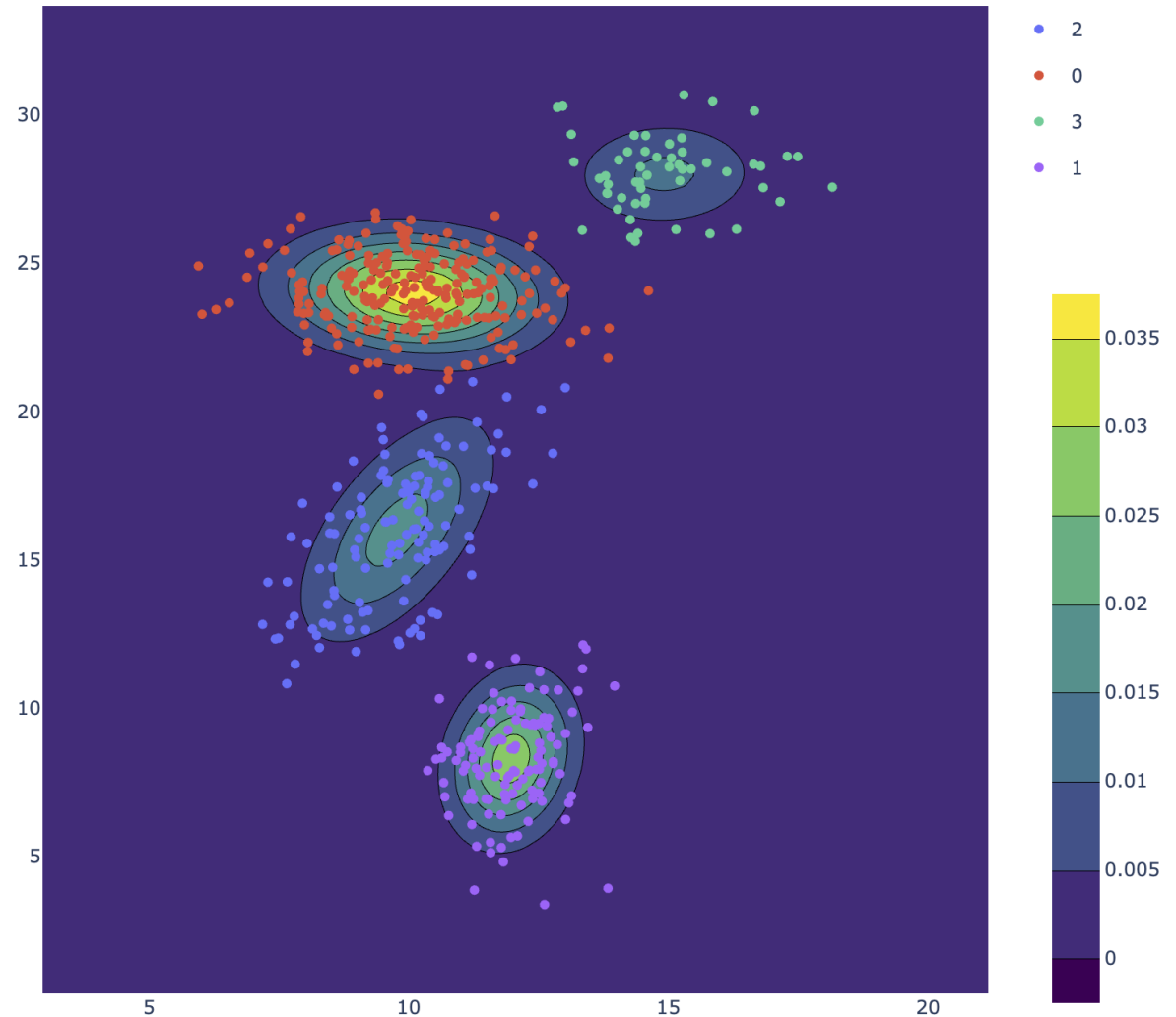
- **Example: $K=3$**



Demo

Gaussian Mixture Model

```
from sklearn.mixture import GaussianMixture
# Create a Gaussian Mixture Model with 4 components
gmm = GaussianMixture(n_components=4, random_state=42, )
# Fit the model to the data
gmm.fit(bikes[['Speed', 'Length']])
# Get the cluster labels
bikes['scikit gmm'] = gmm.predict(bikes[['Speed', 'Length']]).astype(str)
```



The GMM is a **Latent Variable Model**

We introduce a **latent** (unobserved) **assignment variable** z for each $\mathbf{x}$.

$$p(z) = \pi_z$$

We then define the likelihood of $\mathbf{x}$ given z as:

$$p(\mathbf{x} | z) = \mathcal{N}(\mathbf{x} | \mu_z, \Sigma_z)$$

The joint probability of the $\mathbf{x}$ and z is then:

$$p(\mathbf{x}, z) = p(\mathbf{x} | z)p(z) = \mathcal{N}(\mathbf{x} | \mu_z, \Sigma_z)\pi_z$$

If we **marginalize over z** we obtain the **GMM**:

$$p(\mathbf{x}) = \sum_{k=1}^K p(\mathbf{x}, z = k) = \sum_{k=1}^K \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)\pi_k$$

The GMM is a Generative Model

A **generative model** is a model that can **generate the data**.

- The GMM is a generative model.

Recall, the joint distribution has the following factorization:

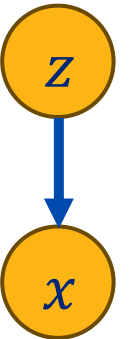
$$p(\mathbf{x}, z) = p(z) p(\mathbf{x} | z)$$

We can draw a **graph** where the parents of a variable are the variables in its **conditional distribution**.

- This is a **graphical model** and is used to represent the **conditional structure** of a probability model.

We can sample data from a graphical model using **ancestor sampling**.

- Sample all the variables that have no parents: $p(z)$
- Continue sampling variables after all their parents have been sampled: $p(\mathbf{x} | z)$



Demo

Sampling from a GMM

```
# Ancestor Sampling to create a synthetic dataset
np.random.seed(42)
N = 100
```

```
mu = np.array([-1, 2, 5])
pi = np.array([0.2, 0.5, 0.3])
Sigma = np.array([0.2, 0.5, .1])
```

✓ 0.0s

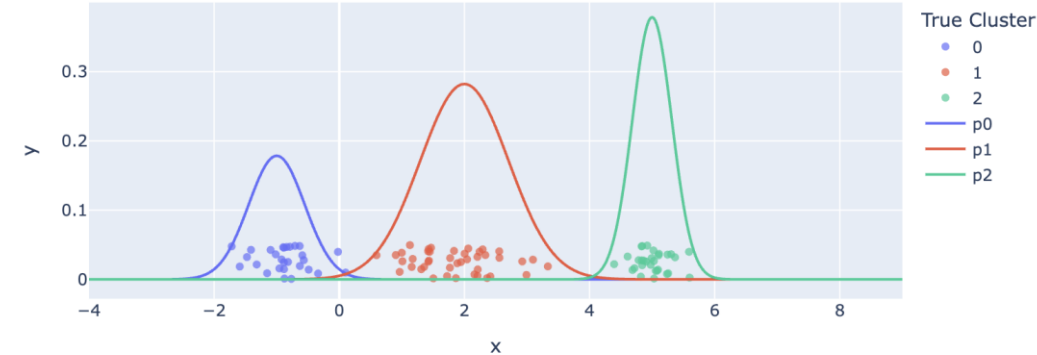
```
z = np.random.choice(len(mu), size=N, p=pi)
x = np.random.normal(mu[z], np.sqrt(Sigma[z]))
```

✓ 0.0s

```
log_likelihood = np.sum(np.log(np.sum(
    pi[z] * norm.pdf(x[:, None], loc=mu[z], scale=np.sqrt(Sigma[z])),
    axis=1 )))
```

✓ 0.0s

Synthetic Dataset from GMM (Log Likelihood: 181.82)



Latent Variable Posteriors

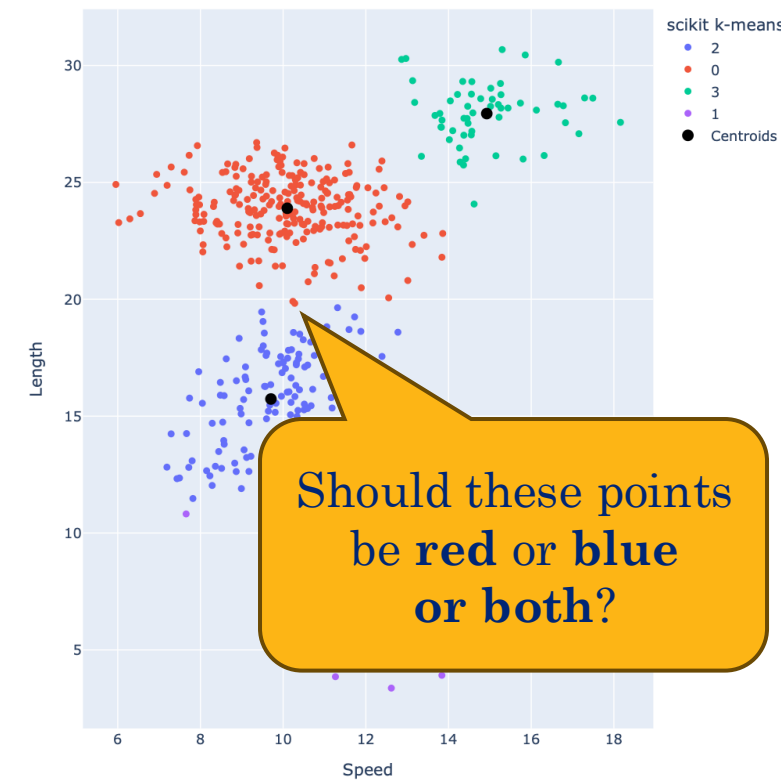
We are interested in modeling the **distribution** (*the uncertainty*) over the **cluster assignment** z .

- **Prior:** $p(z) = \pi_z$
- **Likelihood:** $p(\mathbf{x} | z) = \mathcal{N}(\mathbf{x} | \mu_z, \Sigma_z)$

We can use Bayes rule to compute the posterior:

$$p(z | \mathbf{x}, \pi, \{\mu_k, \Sigma_k\}_{k=1}^K) = \frac{p(\mathbf{x} | z)p(z)}{\sum_{k=1}^K p(\mathbf{x} | z = k)p(z = k)} = \frac{\mathcal{N}(\mathbf{x} | \mu_z, \Sigma_z)\pi_z}{\sum_{k=1}^K \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)\pi_k}$$

If we **could estimate** $\pi, \{\mu_k, \Sigma_k\}_{k=1}^K$ **from the data** we could compute this distribution over cluster assignments.



Estimating the GMM Parameters

We could try to use the method of **maximum likelihood**:

$$\ln \left(p(\{\mathbf{x}_n\}_{n=1}^N \mid \pi, \{\mu_k, \Sigma_k\}_{k=1}^K) \right) = \sum_{n=1}^N \ln \left(\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n \mid \mu_k, \Sigma_k) \right)$$

- however, the summation inside of the log couples the parameters.

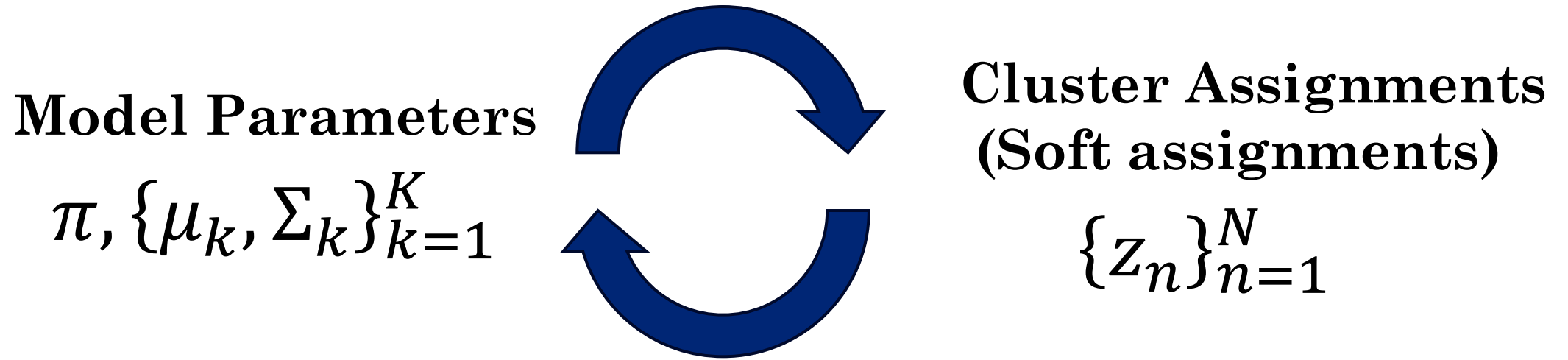
But, if we extend the model to include the **latent variable** z (This is an idealistic case, assuming that $\{z_n\}_{n=1}^N$ have been known/observed, just like $\mathbf{x}_n$.)

$$\ln \left(p(\{\mathbf{x}_n, z_n\}_{n=1}^N \mid \pi, \{\mu_k, \Sigma_k\}_{k=1}^K) \right) = \sum_{n=1}^N \left(\ln(\pi_{z_n}) + \ln \left(\mathcal{N}(\mathbf{x}_n \mid \mu_{z_n}, \Sigma_{z_n}) \right) \right)$$

- We can optimize the parameters directly (sum of log terms).
- But we don't know the values of $\{z_n\}_{n=1}^N$

Quick Recap

If we knew the model parameters, we could easily compute the cluster assignments using the posterior about z .



If we knew the cluster assignments, we could easily estimate the model parameters using maximum likelihood method.

How can we solve this cyclic dependency?

The EM Algorithm

The **Expectation-Maximization (EM)** algorithm is a standard technique for fitting **latent variable** models like the Gaussian Mixture Model.

1. **Initialize** the model parameters θ .
2. **Iterate** until convergence.
 - **E-Step:** Compute the posterior and expected log-likelihood.

$$Q(\theta') = \sum_Z \ln(p(X, Z | \theta')) p(Z | X, \theta)$$

- **M-Step:** Maximize the expected log-likelihood.

$$\theta = \arg \max_{\theta'} Q(\theta')$$

Easy to optimize
joint probability

Current posterior
distribution
over the latent Z

Updates distribution over Z .

The EM Algorithm: **E**-step

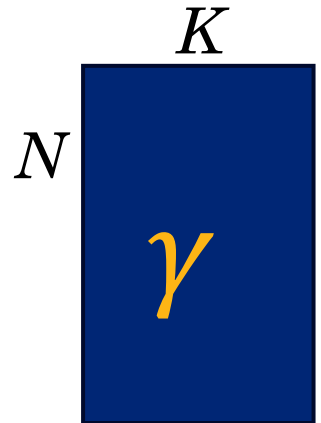
E-Step: We don't know the value of $\{z_n\}_{n=1}^N$ but we can compute posterior

$$\gamma_{nk} = p(z_n = k | \mathbf{x}_n, \pi, \{\mu_k, \Sigma_k\}_{k=1}^K) = \frac{p(\mathbf{x}_n | z_n) p(z_n)}{\sum_{k'=1}^K p(\mathbf{x}_n | k') p(k')} = \frac{\mathcal{N}(\mathbf{x}_n | \mu_{z_n}, \Sigma_{z_n}) \pi_{z_n}}{\sum_{k'=1}^K \mathcal{N}(\mathbf{x}_n | \mu_{k'}, \Sigma_{k'}) \pi_{k'}}$$

- γ_{nk} are “soft” assignments to each class.

Using the γ_{nk} we can compute the **expected log likelihood**:

$$\begin{aligned} Q(\pi, \{\mu_k, \Sigma_k\}_{k=1}^K) &= \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} \ln \left(p(\mathbf{x}_n, z_n = k | \pi, \{\mu_k, \Sigma_k\}_{k=1}^K) \right) \\ &= \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\ln(\pi_k) + \ln(\mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k))) \end{aligned}$$



The EM Algorithm: M-Step (π_k)

M-Step: Compute the parameters that maximize Q :

$$Q(\pi, \{\mu_k, \Sigma_k\}_{k=1}^K) = \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\ln(\pi_k) + \ln(\mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)))$$

Lagrangian for the
normalization constraint

Optimizing with respect to π_k :

$$\frac{\partial}{\partial \pi_k} \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\ln(\pi_k) + \ln(\mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k))) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right) = \sum_{n=1}^N \frac{\gamma_{nk}}{\pi_k} + \lambda = 0$$

$$1 = \sum_{k=1}^K \pi_k = -\frac{1}{\lambda} \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} = -\frac{1}{\lambda} \sum_{n=1}^N 1$$
$$= -\frac{N}{\lambda} \Rightarrow \lambda = -N$$

$$\pi_k = -\frac{1}{\lambda} \sum_{n=1}^N \gamma_{nk}$$
$$\Rightarrow \frac{1}{N} \sum_{n=1}^N \gamma_{nk} = \frac{N_k}{N}$$

$$N_k = \sum_{n=1}^N \gamma_{nk}$$

The EM Algorithm: **M-Step** (μ_k)

M-Step: Compute the parameters that maximize Q :

$$Q(\pi, \{\mu_k, \Sigma_k\}_{k=1}^K) = \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\ln(\pi_k) + \ln(\mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)))$$

Optimizing with respect to μ_k :

$$\frac{\partial}{\partial \mu_k} \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\ln(\pi_k) + \ln(\mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k))) = \sum_{n=1}^N \gamma_{nk} \Sigma_k^{-1} (\mathbf{x}_n - \mu_k) = 0$$

$$\mu_k = \frac{1}{\sum_{n=1}^N \gamma_{nk}} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n$$

The EM Algorithm: **M-Step** (Σ_k)

M-Step: Compute the parameters that maximize Q :

$$Q(\pi, \{\mu_k, \Sigma_k\}_{k=1}^K) = \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\ln(\pi_k) + \ln(\mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)))$$

Optimizing with respect to Σ_k requires matrix calculus (out of scope):

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} (\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T$$

Summarizing the other parameters:

$$N_k = \sum_{n=1}^N \gamma_{nk}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n$$

The EM Algorithm for GMMs

1. **Initialize** the model parameters $\pi, \{\mu_k, \Sigma_k\}_{k=1}^K$ (often using k-means + density estimation)
2. **Iterate** until convergence

- **E-Step:** compute posterior and expected log-likelihood

$$\gamma_{nk} = \frac{\mathcal{N}(\mathbf{x}_n | \mu_{z_n}, \Sigma_{z_n}) \pi_{z_n}}{\sum_{k'=1}^K \mathcal{N}(\mathbf{x}_n | \mu_{k'}, \Sigma_{k'}) \pi_{k'}}$$

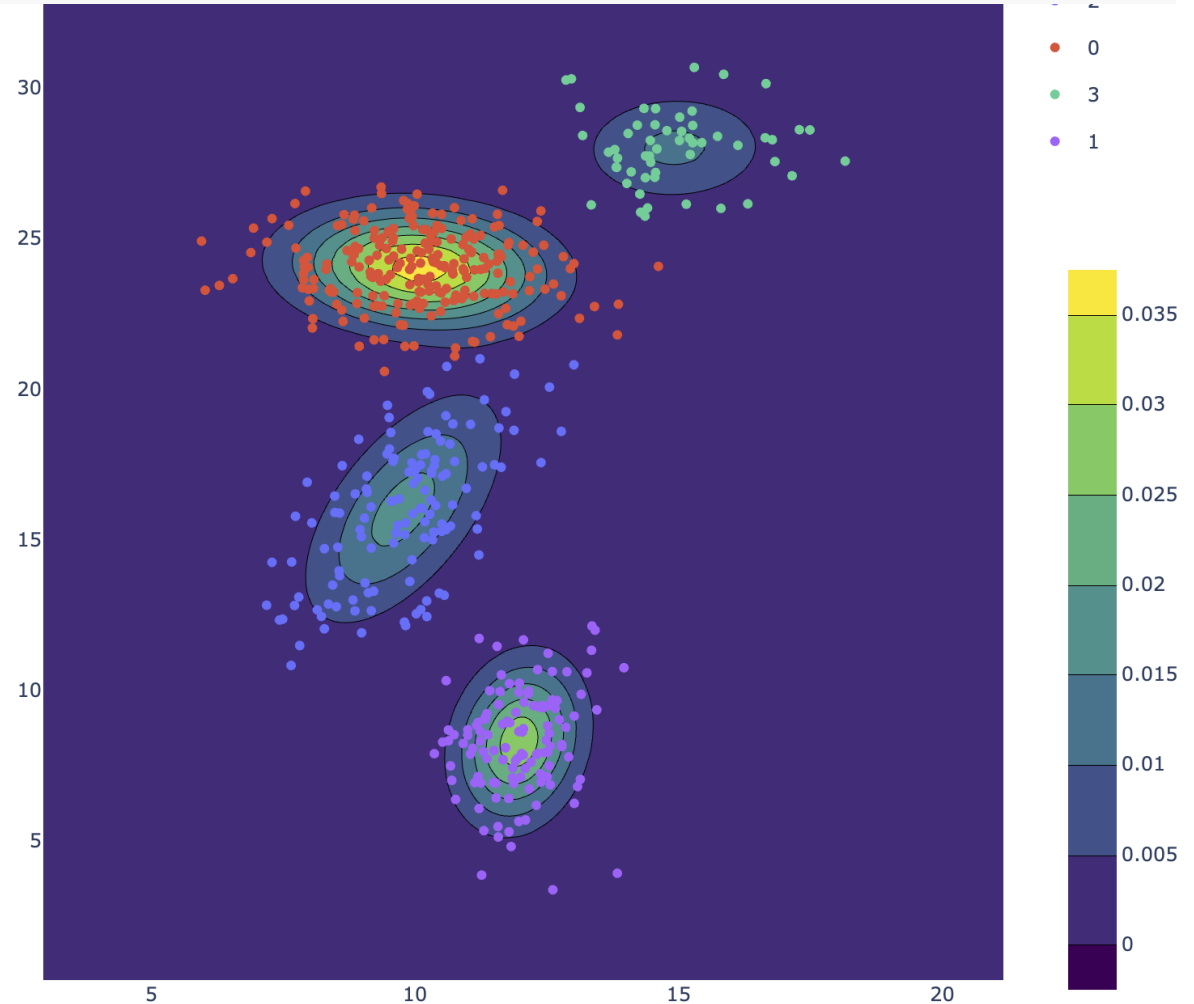
- **M-Step:** maximize the expected log-likelihood

$$N_k = \sum_{n=1}^N \gamma_{nk}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n$$
$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} (\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T$$

Demo

Implement GMMs

```
from sklearn.mixture import GaussianMixture
# Create a Gaussian Mixture Model with 4 components
gmm = GaussianMixture(n_components=4, random_state=42, )
# Fit the model to the data
gmm.fit(bikes[['Speed', 'Length']])
# Get the cluster labels
bikes['scikit gmm'] = gmm.predict(bikes[['Speed', 'Length']]).astype(str)
```



Lecture 4

Density Estimation and Gaussian Mixture Models

Reference Book Chapters:

- **Probability:** Chapter 2.[1-2], 2.3.[1-3] (we will return to 2.4 onward later)
- **Distributions:** Chapter 3.1, 3.2.1 (the rest of 3 is more advanced Gaussians)
- **Clustering:** Chapter 15.1 (k-means), 15.2 (Gaussian Mixture Models)